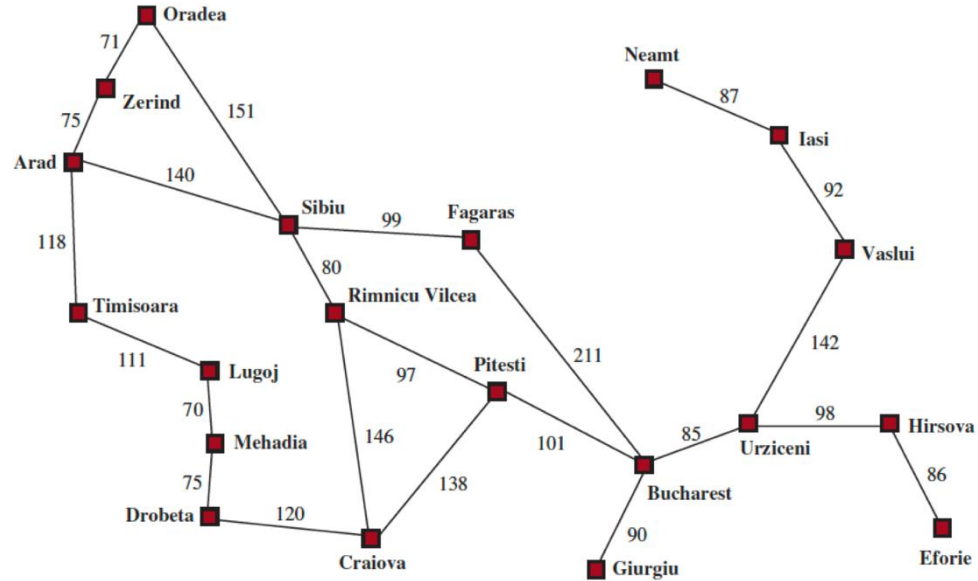


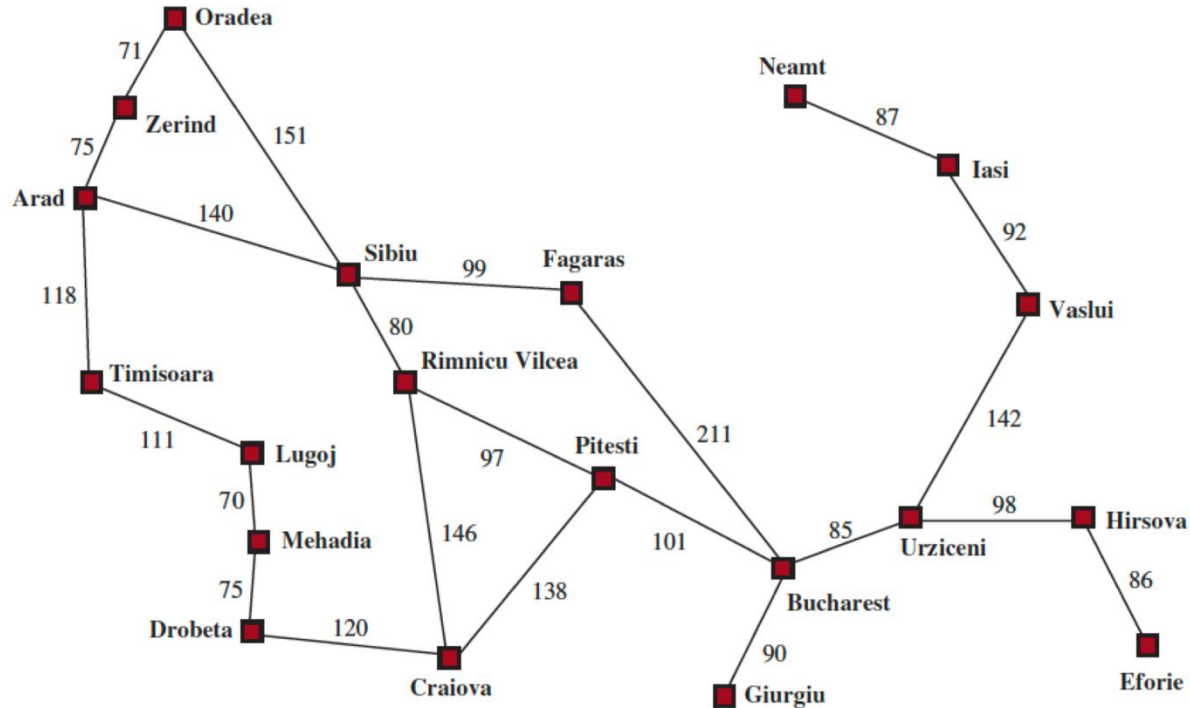
Breadth First Search for Travelling Salesman problem

Week 9 Laboratory · 2 hours



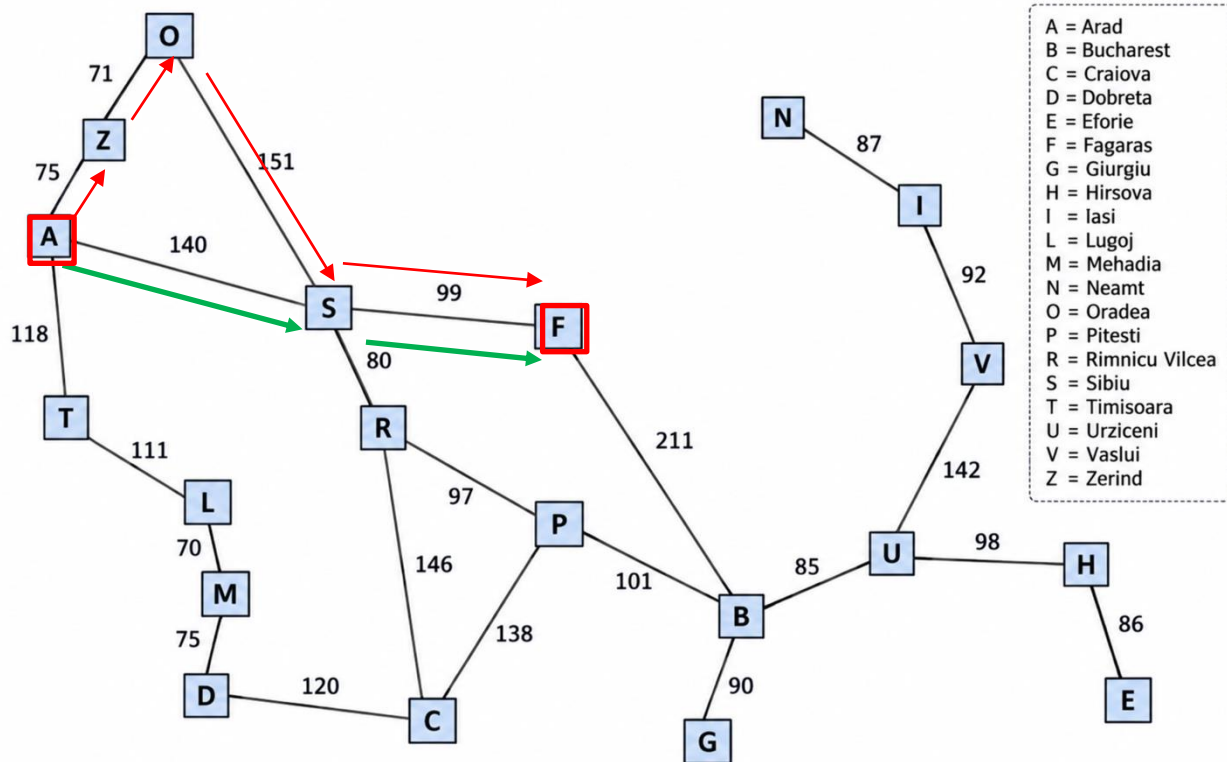
Given a map Romania

A simplified map between cities



Given a map Romania

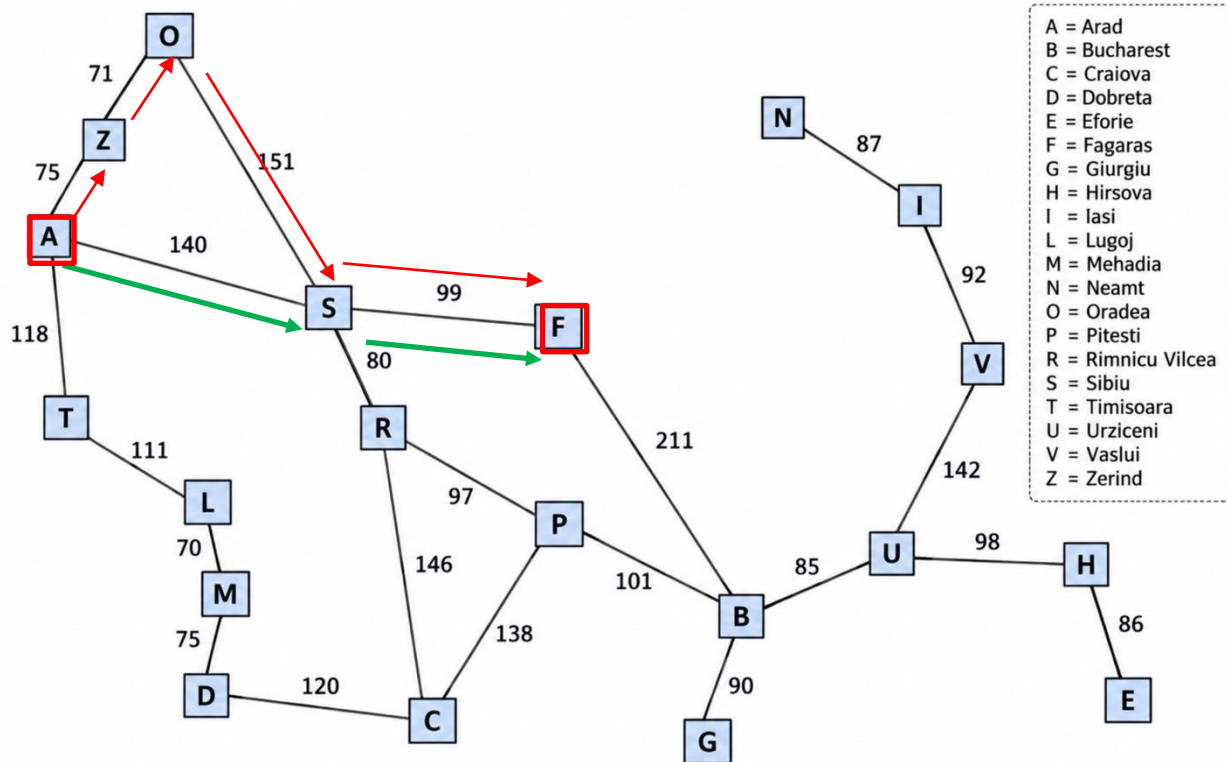
A simplified map between cities



How can we travel from a city to another?

What are the routes from A → F?

What is the shortest route from A → F?



Start: A
Goal: F

Search Algorithms

Task 2 – Solving Problems by Searching (5 marks): Deadline 16 Oct 2026

You need to implement 2 types of search algorithms:

Uninformed search

1. **Breadth-First Search**
2. Depth-First Search,
3. Uniform-cost search.

Informed search

1. Best-First Search,
2. Greedy Best-First Search
3. A* search.

Today's Lab Work

We need to implement Breadth-First Search:

Uninformed search

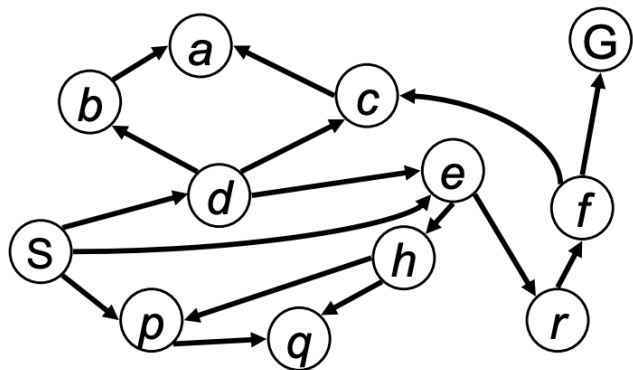
1. **Breadth-First Search**
2. Depth-First Search,
3. Uniform-cost search.

Informed search

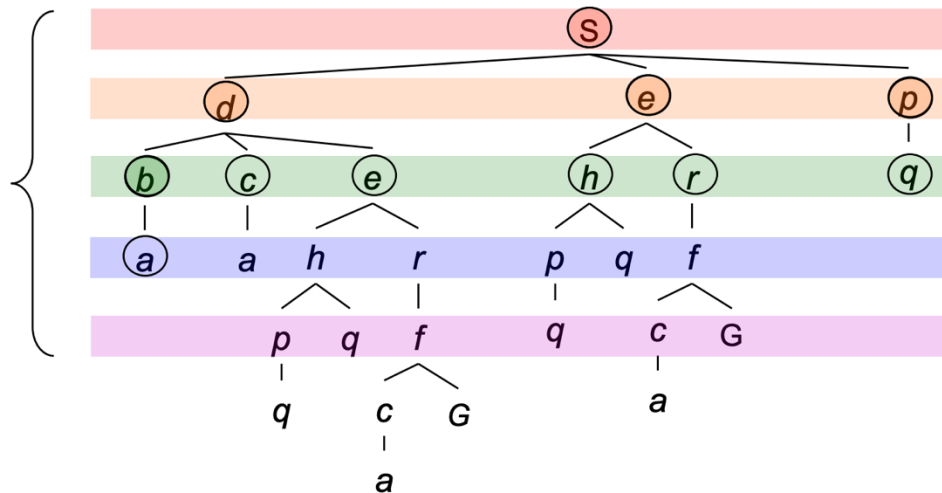
1. Best-First Search,
2. Greedy Best-First Search
3. A* search.

Review: Breadth-First Search

Visit the shallowest nodes first!



Search
Tiers



Implementation: Frontier is a FIFO (First in, First out) queue

What you can do today ?

We need to implement Breadth-First Search:

Interestingly, we **can do all of the Task 2 experiments!**

Uninformed search

1. Breadth-First Search
2. Depth-First Search,
3. Uniform-cost search.

Informed search

1. Best-First Search,
2. Greedy Best-First Search
3. A* search.

What you can do today ?

We need to implement Breadth-First Search:

Interestingly, we **can do all of the Task 2** experiments!

Task 2 – Solving Problems by Searching (5 marks)

Specifications

Searching is an important technique used by AI agents to solve a wide range of problems, including the travelling salesman problem, the 8-puzzle problem, the N-queens problem, and other practical problems that can be formulated as search tasks in a state space. In this lab task, you are required to implement two types of search methods and their specific algorithms:

- **Uninformed** search: Breadth-first search, Depth-first search, and Uniform-cost search.
- **Informed** search: Best-first search, Greedy best-first search, and A* search.

ID	Algorithm	Explored states	Solution	Path cost	Time (seconds)
1					
2					
3					
4					
5					

What I really want you to do today ?

Do teamwork on your group project.
Discuss with your teammates!